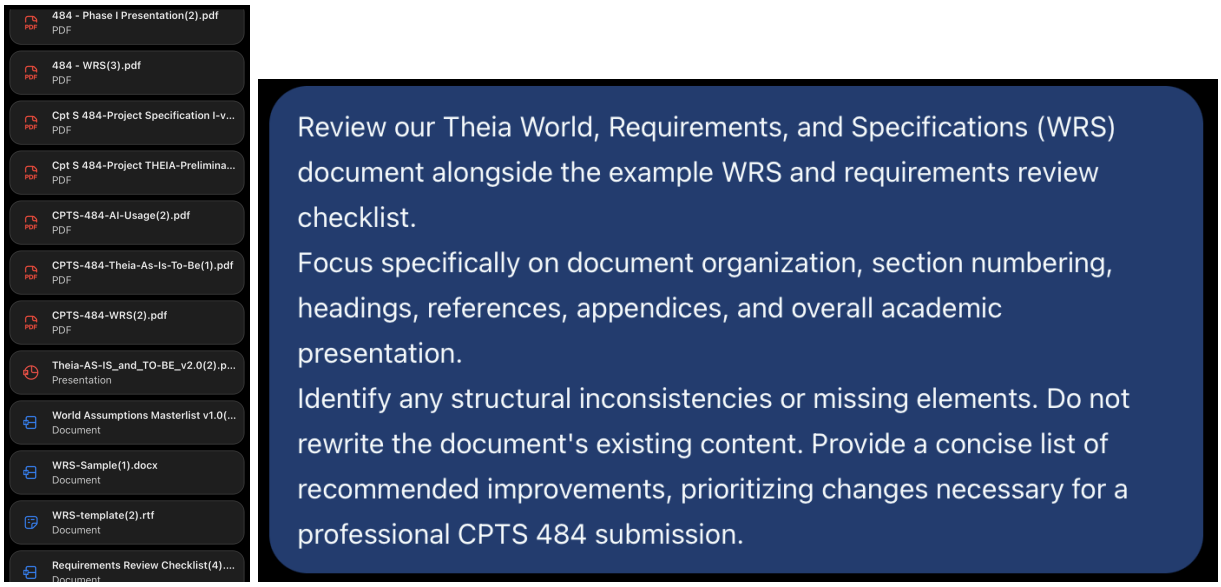
Disclosure of AI Usage

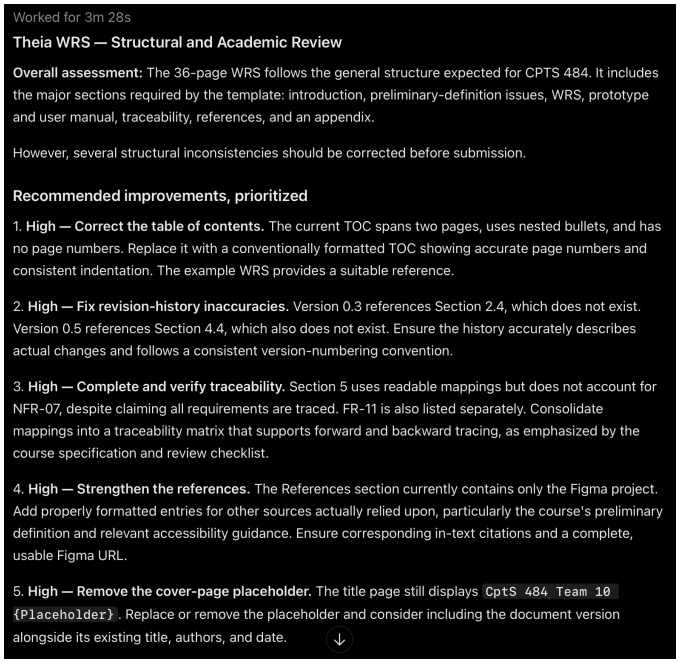
Alex

I used ChatGPT to cross-check our work against the submission requirements, refine spelling and grammar, format the reference to Seth's Figma project, and organize the WRS document's table of contents. All AI-generated suggestions were reviewed using my own judgment to determine which changes were necessary and which were unnecessary, irrelevant, or superfluous.

Prompt:



Response:



Prompt:

Review Section 4.1, Prototype Overview, of our WRS document.
The section currently links directly to our Figma interface mockups:
https://www.figma.com/design/nUloE2z6xJFQULTEBOEz66/Theia---Interface-Mockups?node-id=0-1
Would it be more appropriate to cite the mockups as a numbered reference rather than include a standalone hyperlink?
Recommend the correct academic citation approach consistent with the example WRS. Keep changes minimal and preserve the existing wording wherever possible.

Response:

Worked for 23s

Yes. A numbered reference is more appropriate for an academic WRS, particularly because the example WRS uses numerical citations such as `[1]` and `[2]` that correspond to entries in its References section.

However, **your current WRS already implements this correctly.** No substantive changes are necessary.

1. Section 4.1 — Prototype Overview

Keep the existing final sentence:

> The complete interface mockups are available in the project's Figma design [1].

This is appropriately concise and consistent with the example WRS.

2. References section

Retain the existing reference entry, ensuring the full URL is clickable:

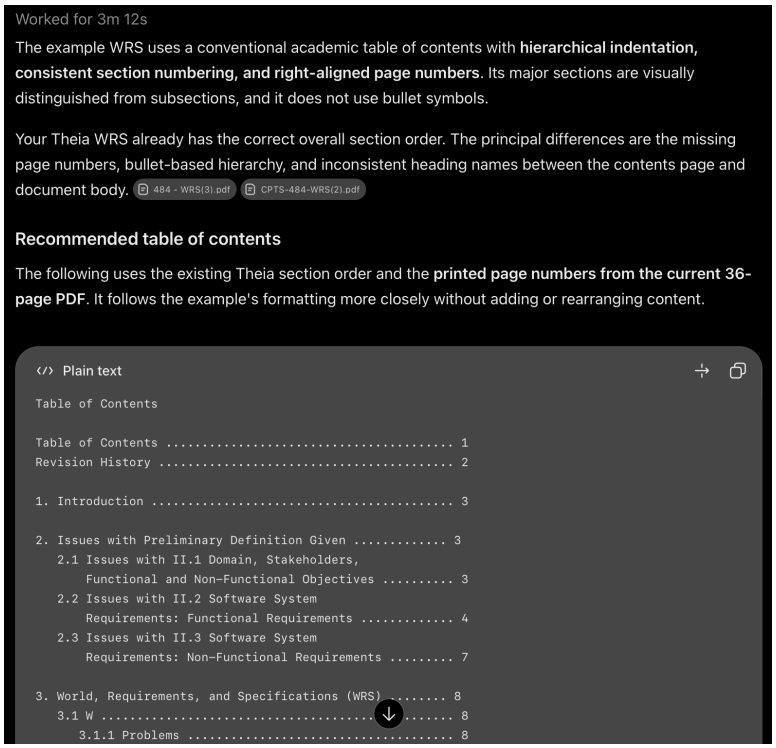
> [1] Theia Project Team, "Theia – Interface Mockups," Figma, 2026. [Online]. Available: https://www.figma.com/design/nUloE2z6xJFQULTEBOEz66/Theia---Interface-Mockups?node-id=0-1 . Accessed: Oct. 18, 2026.

This follows an IEEE-style numbered reference format, which is appropriate for software engineering documentation.

Prompt:

Examine the table of contents in the provided example WRS document.
Compare its formatting, heading hierarchy, numbering conventions, indentation, and page-number presentation with the structure of our Theia WRS.
Recommend a similar table of contents that fits our actual document.
Preserve our existing section order and distinguish between required structural changes and optional formatting improvements.

Response:



Worked for 3m 12s

The example WRS uses a conventional academic table of contents with **hierarchical indentation, consistent section numbering, and right-aligned page numbers**. Its major sections are visually distinguished from subsections, and it does not use bullet symbols.

Your Theia WRS already has the correct overall section order. The principal differences are the missing page numbers, bullet-based hierarchy, and inconsistent heading names between the contents page and document body. 484 - WRS(3).pdf CPTS-484-WRS(2).pdf

Recommended table of contents

The following uses the existing Theia section order and the **printed page numbers from the current 36-page PDF**. It follows the example's formatting more closely without adding or rearranging content.

Plain text

```
Table of Contents

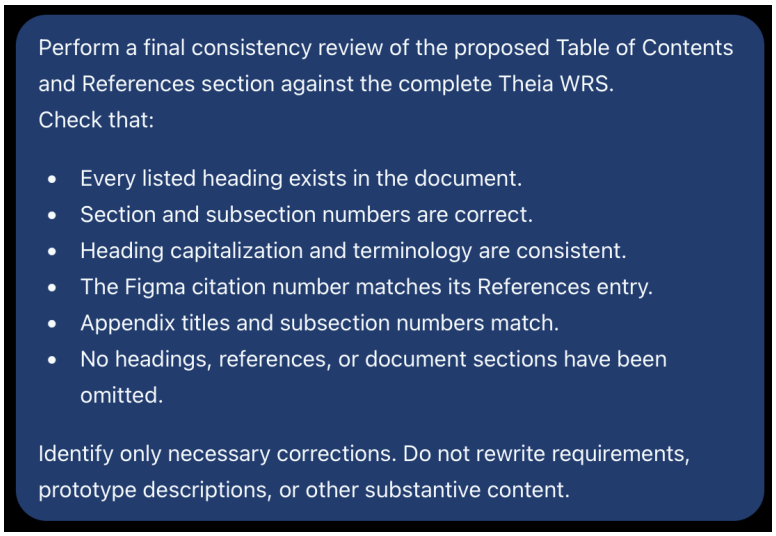
Table of Contents ..................................... 1
Revision History ...................................... 2

1. Introduction ........................................ 3

2. Issues with Preliminary Definition Given ............. 3
   2.1 Issues with II.1 Domain, Stakeholders,
       Functional and Non-Functional Objectives ......... 3
   2.2 Issues with II.2 Software System
       Requirements: Functional Requirements ............ 4
   2.3 Issues with II.3 Software System
       Requirements: Non-Functional Requirements ........ 7

3. World, Requirements, and Specifications (WRS) ....... 8
   3.1 W ..................................... ..... 8
      3.1.1 Problems ..................................... 8
```

Prompt:



Perform a final consistency review of the proposed Table of Contents and References section against the complete Theia WRS.
Check that:

- Every listed heading exists in the document.
- Section and subsection numbers are correct.
- Heading capitalization and terminology are consistent.
- The Figma citation number matches its References entry.
- Appendix titles and subsection numbers match.
- No headings, references, or document sections have been omitted.

Identify only necessary corrections. Do not rewrite requirements, prototype descriptions, or other substantive content.

Response:

Worked for 2m 37s

The proposed Table of Contents is **structurally accurate overall**, and the Figma citation and appendix numbering are correct. Three corrections are needed to achieve complete consistency with the 36-page Theia WRS.

1. Add the omitted Section 2 headings

The proposed TOC includes third-level headings under Section 4 but omits the 16 numbered issue headings under Section 2. For a complete heading inventory, add:

Heading	Printed page
2.1.1 Meaning of significant location	3
2.1.2 Target-user variation	4
2.1.3 Meaning of emergency dial-out	4
2.1.4 Definition of accessibility success	4
2.2.1 Current-location update and beacon interpretation	4
2.2.2 Destination selection and route availability	5
2.2.3 Route instruction content	5
2.2.4 Emergency dial-out workflow	5
2.2.5 Vertical navigation and floor transitions	6
2.2.6 Static obstacle identification	6
2.2.7 Fall detection and emergency escalation	6

Prompt:

Based on the completed review, specify exactly where the Table of Contents and References should appear in the final Theia WRS. Recommend a professional academic layout consistent with the example, including title page, table of contents, revision history, main sections, references, and appendices.
Explain how to apply Google Docs heading styles and generate an automatic Table of Contents with right-aligned page numbers.
Keep the recommendations concise and avoid changes unrelated to document formatting.

Response:

Worked for 46s

For the final Theia WRS, use the same overall organization as the example WRS: **title page → table of contents → revision history → main document → references → appendices**.

1. Recommended document layout

Order	Section	Placement
1	Title Page	First page, preferably without a visible page number
2	Table of Contents	Immediately after the title page, starting on a new page
3	Revision History	Immediately after the completed TOC, starting on a new page
4	Sections 1–5	Introduction through Traceability, in their existing order
5	References	Immediately after Section 5, preferably on a new page
6	Appendix A: Phase I Plan Revisions	After References, starting on a new page

This follows the example's front-matter arrangement and preserves the Theia document's existing content.

2. Apply Google Docs heading styles

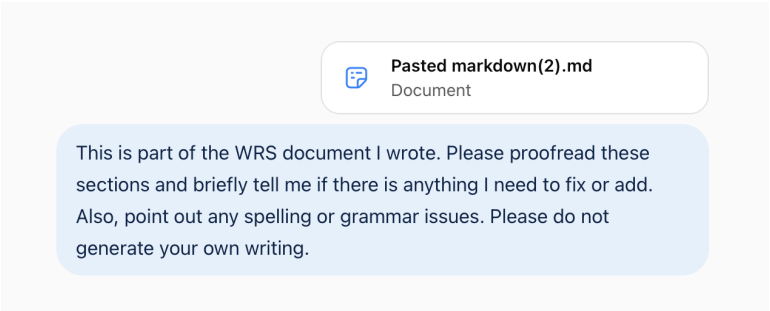
Use actual heading styles rather than manually bolding text.

Document element	Google Docs style
Document title on cover	Title
Sections 1–5	Heading 1
Sections 2.1, 3.1, 4.1, etc.	Heading 2
Sections 2.1.1, 3.1.1, 4.1.1, etc.	Heading 3

Danbi

I used AI to refine my writing (mainly spelling and grammar). I also checked for suggestions and revised the WRS document accordingly.

Prompt:



Response:

Overall assessment

The WRS sections are generally well organized and understandable. The structure of **issue description → options → provisional decision and rationale** is consistent, and most decisions are specific enough to guide the Phase I prototype.

The main things to fix are **consistency between requirements, scope clarification, and a few grammar issues**. I would address the following before submitting.

1. Things to fix or add

Priority	Section	Issue
High	2.1.3, 2.2.4, 2.2.7	Emergency handling is described in three places. The behavior is mostly consistent, but make sure all three specify the same trigger, 30-second countdown, and native Phone app handoff.
High	2.2.5, 2.2.6	Single-floor navigation and static obstacle guidance need to be reconciled. Clarify whether the pillar guidance is part of a route on the single-floor prototype map.
High	2.3.5	The requirement to instruct the user to stop walking is a safety-critical behavior. Consider specifying how the app communicates this accessibly, such as through spoken instructions, and what happens if location cannot be restored.
Medium	2.1.4, 2.3.1	Accessibility checks overlap. This is reasonable, but ensure the checks are measurable and testable rather than just general goals.
Medium	2.3.2	The five-second response-time target is useful. Clarify whether it applies to all simulated events and user actions, including route updates and emergency alerts, or only ordinary interactions.
Medium	2.3.4	The target platform is identified, but the minimum iOS version and supported test environment remain unspecified. Decide whether these details are necessary for your prototype.
Medium	2.3.3	“Will not transmit location or data to a server” is clear, but consider whether the prototype stores any data locally and whether that data persists after the app closes.

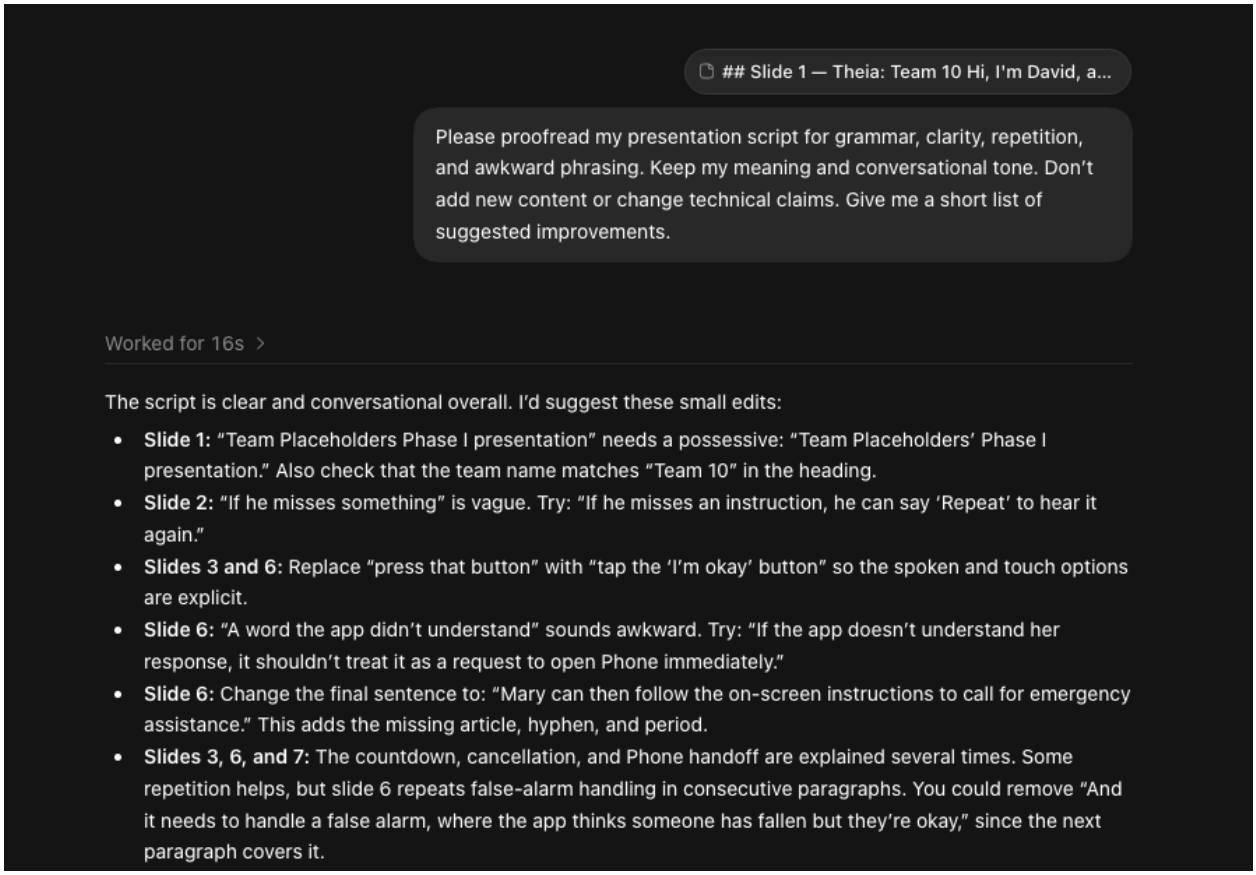
2. Spelling and grammar issues

I did not find major spelling errors. These are the specific wording issues I would correct.

Section	Issue	What to fix
2.1.1	“Include enumerated set of rooms and shared destinations.”	Add the article **“an”** before “enumerated set.”
2.1.1	“entrances, exits, stairs, elevators, restrooms, and so on”	“And so on” is vague for a requirements document. Consider defining the intended categories more precisely.
2.2.1	“or repeated signals cause repeated announcements”	The grammatical structure is inconsistent with the preceding phrases. Use **“whether repeated signals cause repeated announcements.”**
2.3.4	“Target an iPhone/iOS interaction model and will be evaluated…”	The sentence has a parallel-structure problem. Align the two clauses grammatically.
2.3.4	“iPhone/iOS”	Slightly redundant terminology. Consider using one consistent term for the target platform.
2.3 heading	“Issues with II.3 software system requirements: Non-functional requirements”	Capitalization differs from the preceding headings. Standardize heading capitalization.

David

I used AI to help proofread my writing.



I used OpenAI Codex to proofread my presentation script for grammar, clarity, repetition, and awkward phrasing. After reviewing its suggestions against my intended meaning and the project materials, I chose to keep my original wording because it was clear and suited my speaking style.

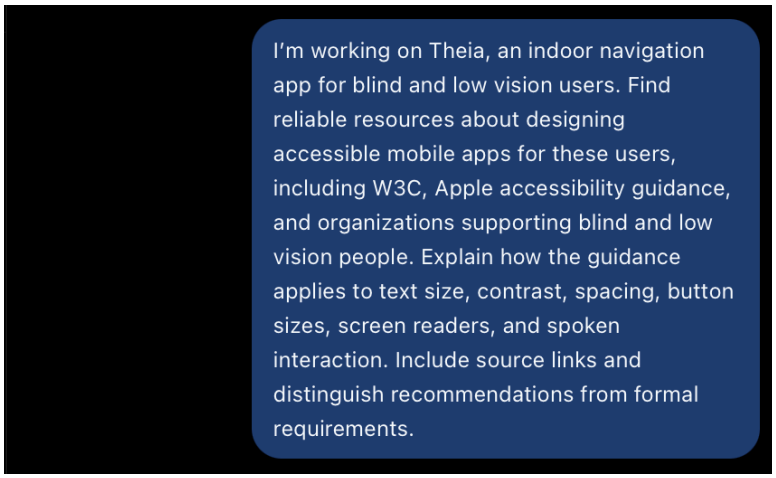
Johanna

N/A.

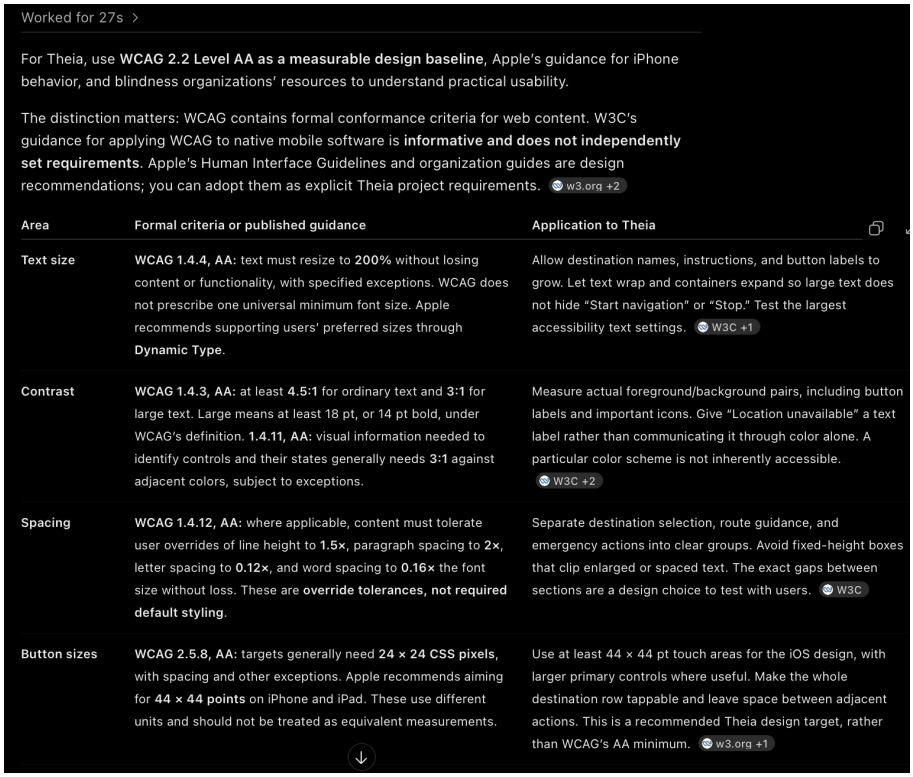
Seth

I used AI to help research interface design practices for blind and low vision users, including font size, text spacing, color contrast, control size, and video interaction. I used accessibility guidance found online to inform the Figma prototypes and reviewed the designs against those recommendations. I also used AI to check my WRS section for spelling and grammar. I then reviewed the suggestions and revised my section of the WRS.

Prompt:



Response:



Screen readers	**WCAG 4.1.2, A:** controls need programmatically available names, roles, and relevant states/values. **4.1.3, AA:** qualifying status messages must be available to assistive technology without receiving focus. Apple and RNIB also emphasize meaningful labels and logical navigation.	Label controls “Choose destination” and “Repeat instruction,” rather than leaving microphone or speaker icons unnamed. Expose selection states, provide a sensible reading order, and announce “Route ready” or “Location unavailable” without unexpectedly moving focus. W3C +3

Spoken interaction needs its own design decisions. VoiceOver reads and operates the interface; Theia’s spoken route instructions and speech recognition are separate features. RNIB explains why clear spoken labels, sensible reading order, and completion feedback matter to users navigating one element at a time. rnib.org.uk

Based on that guidance, I recommend these Theia behaviors. They are **project recommendations**, rather than specific WCAG mandates:

- Give concise instructions, such as “Turn left at the next hallway,” and keep the current instruction available as accessible text.
- Provide accessible **Repeat**, **Pause**, and **Stop navigation** controls.
- Coordinate route speech with VoiceOver so competing announcements do not obscure instructions.
- Confirm recognized destinations: “You selected Room 204. Start navigation?”
- Offer destination selection and text entry alongside voice input.
- Clearly announce lost positioning instead of continuing to present an outdated instruction as current.

These resources are good starting points:

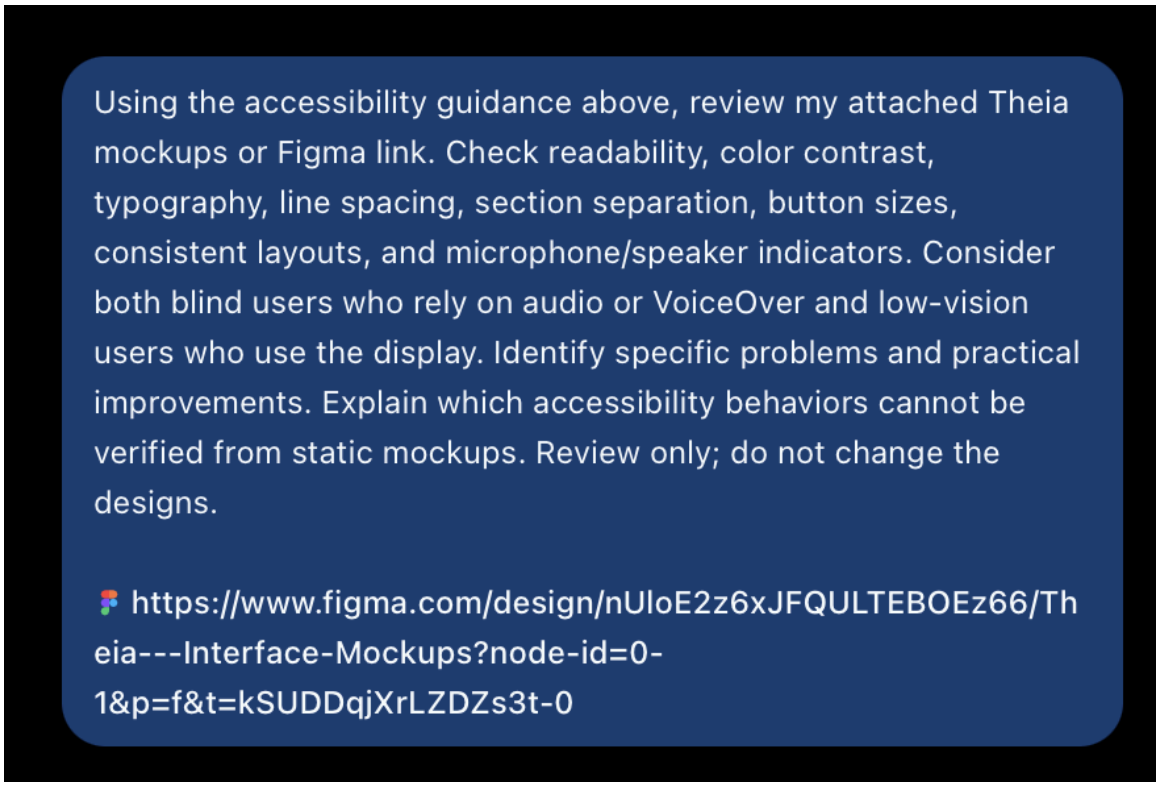
Resource	Why it is useful
W3C WCAG 2.2	The authoritative source for criterion wording, levels, exceptions, and conformance requirements.
W3C WCAG2ICT and WCAG2Mobile	Explain how WCAG applies to software and mobile apps; both provide informative guidance.
Apple Accessibility HIG, Typography, and	Platform ance for adaptable presentation and accessible

These resources are good starting points:

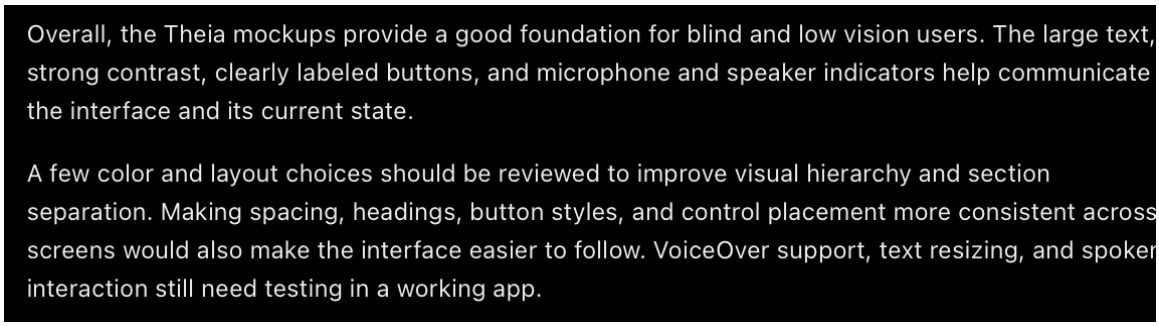
Resource	Why it is useful
W3C WCAG 2.2	The authoritative source for criterion wording, levels, exceptions, and conformance requirements.
W3C WCAG2ICT and WCAG2Mobile	Explain how WCAG applies to software and mobile apps; both provide informative guidance.
Apple Accessibility HIG, Typography, and VoiceOver	Platform guidance for adaptable presentation and accessible iPhone interaction.
RNIB App Guidelines, PDF	Practical guidance on labels, headings, focus, gestures, magnification, and operating-system preferences.
American Foundation for the Blind: Access Barriers in Websites and Mobile Apps	User research documenting problems including unlabeled controls, small text, poor contrast, clutter, and unexpected focus changes.

For validation, test complete journeys with blind and low vision participants using their preferred settings and assistive technology. AFB’s findings show why visual review alone is insufficient. For Theia, include choosing a destination, following instructions, repeating guidance, and recovering fro lost location. Figma mockups can demonstrate layout and intended behavior; working software is needed to verify screen-reader operation and speech coordination. American Foundat...

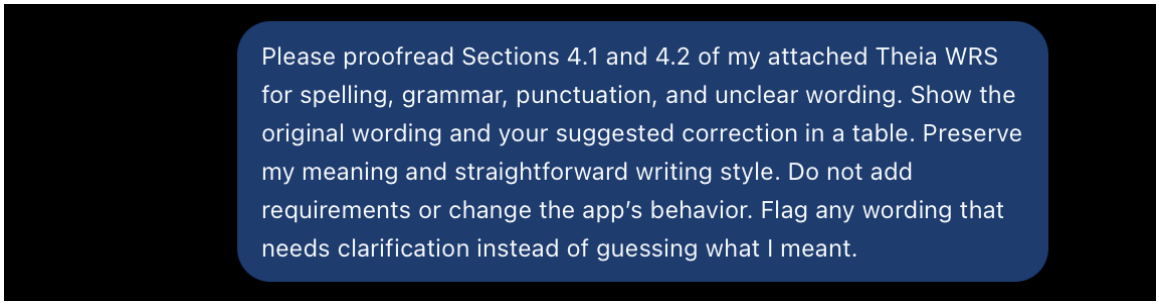
Prompt:



Response:



Prompt:



Response:

Item	What to fix
Figma link	Fill in the blank after “Link to Figma Design.”
Missing command	Add **“Speak destination”** to 4.2 because the Home screen and manual use it.
Manual emergency assistance	Clarify whether you removed only its separate mockup or the functionality. Your presentation still includes user-initiated emergency requests, and your manual still describes the removed prompt.
Presentation consistency	Scenario 1 asks users to provide their current location, while your prototype identifies it through simulated beacons. The presentation also describes automatic dial-out, while your draft describes a native Phone app handoff. Align those descriptions.
Proofreading	Correct wording such as **“Browser Rooms” → “Browse rooms,” “Listending” → “Listening,” “retires” → “retries,”** and **“Im Ok” → “I’m okay.”**